MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RADIATION COUNTER

TYPE 5979

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Mica end window, halogen-filled, high sensitivity, self quenching for detection of beta and gamma radiations.

Dimensions: See figure 1

Mounting position: Any

Weight: 2.82 ounces nominal

ABSOLUTE RATINGS:

Parameter:	Ebb	TA	Rp
Unit:	Vdc	°C	Meg
Maximum:	725	75	---
Minimum:	675	-40	---
TEST CONDITIONS:	700	---	1.0

GENERAL:

Qualification - Required (See note 2)

Marking - See note 3

Holding period - See note 11

Response characteristics - See note 1

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
6201	Background, contamination, and photosensitivity	t = 2 minutes (see note 5)	0.65	II	N/t	---	60	Npm
6201	Photosensitivity	t = 2 minutes (see notes 5 and 6)	0.65	II	N/t	---	20	Npm
6211	Starting voltage	Pulse amplitude = 1 volt; R1 = C1 = C2 = 0; R2 = 1 Meg; C3 = 0.01 μF; Nps = 200 (max)	0.65	II	Es	---	640	Vdc
6216	Relative plateau slope	Voltage range = 660 to 760 Vdc	0.65	II	Ps	---	0.2	%/Vdc
6216	End-of-plateau voltage	Ps = 0.3%/Vdc (see note 14)	0.65	II	Ee	800	---	Vdc
6216	Plateau length	See note 13	0.65	II	Pl	200	---	Vdc
6221	Response count rate and current (gamma)	t = 2 minutes	0.65	II	N/t	85	115	Nps
	Quality conformance inspection, part 2							
6221	Response count rate and current (beta)	t = 2 minutes	---	---	N/t	See note 8	---	---
6221	Response count rate and current	Gamma (radium); rate = 5 mr/hr; Ebb = 710 Vdc	---	---	Ib	---	5	μAdc
6226	Pulse amplitude (1)	Ebb = 675 Vdc; Nps = 200 (max); R1 = 0.9 Meg; R2 = 0.1 Meg; C1 = 330 pF; C2 = 3,000 pF; C3 = 0.01 μF; multiply oscilloscope reading by 10	---	---	eo	2.5	---	v
6226	Pulse amplitude (2)	Ebb = 725 Vdc; Nps = 200 (max); R1 = 0.9 Meg; R2 = 0.1 Meg; C1 = 330 pF; C2 = 3,000 pF; C3 = 0.01 μF; multiply oscilloscope reading by 10	---	---	eo	---	65	v
	Quality conformance inspection, part 3							
---	Life-test provisions	Group A; counting rate = 1,000 Nps (min); t = 500 hours	---	---	---	---	---	---
---	Life-test end point	See note 10	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
(C)	Periodic-check tests							
1031	Variable-frequency vibration	See notes 4 and 12	---	---	---	---	---	---
---	Temperature cycling	See note 9	---	---	---	---	---	---
1041	Shock	Hammer angle - 40° (see notes 4 and 7)						
6205	Leakage current	Ebb = 500 Vdc	---	---	Llb	---	0.5	μAdc

NOTES:

1. The response characteristics, current and count rate versus field intensity, of a typical type 5979 radiation counter tube are shown on figure 3. In determining tube response, Ebb = 700 Vdc and Rp = 1 Meg. Tube current is shown as measured by means of a microammeter in series with the tube cathode, and the count rates as determined using a scaler having a resolving time of 5 microseconds and a discrimination level of 0.25 volts.

2. A minimum of 28 samples shall be submitted for qualification test. The samples to be submitted for qualification testing will be broken down into the following subgroups:

(1) Shelf life only	6
(2) Electrical, including life test	6
(3) Shock test only	10
(4) Variable-frequency vibration	6

3. Each tube is to have an individual serial number permanently marked on the cathode.

4. Criterion for passing this test shall be compliance after test of at least 80 percent of the tubes with the requirement for:
   (a) Starting voltage
   (b) Response (count rate) (1)
   (c) Pulse amplitude (1) and (2)
   (d) End-of-plateau voltage
   (e) Relative plateau slope

5. In determining tube response (count rate) Ebb = 700 Vdc and Rp = 1 Meg. Tube count rates shall be determined using a scaler having a resolving time of 5 microseconds and a discrimination level of 0.25 volts.

6. The tube end window shall be exposed to radiation from a General Electric 15-watt germicidal lamp and a General Electric 15-watt fluorescent lamp, or equivalents, with relative positions as shown on figure 2.

7. The tube shall be rigidly mounted in position X, Y, and Z by means of a clamp at the center of the cathode. The tube shall be given 10 blows at the X position and 5 blows at each of the other test positions.

8. The counting rate shall not vary by more than ±0.3 N/t from the value of N/t of the calibrated beta source furnished with the standard excitation unit.

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NOTES: -Continued

9. With the tube in a field giving 100 ± 10 Nps at 700 Vdc, determine tube response (count rate) at 675, 700, and 725 Vdc at each of the following temperatures and in the order shown:

   (a) Room temperature
   (b) -40°C
   (c) Room temperature
   (d) 75°C
   (e) Room temperature

   A minimum stabilization time of 30 minutes is to be allowed at each temperature. The absolute count rate at 700 Vdc, and the average relative plateau slope as determined by the readings at the 3 voltages (best average position of straight line), shall not differ from the initial readings at room temperature by more than 10 percent, and ±0.1 percent/volt, respectively, at any of the four subsequent temperatures shown above.

10. The limits for acceptability shall be as follows:

    Photosensitivity: 35 Npm, maximum
    Background and contamination: Initial value + 20 Npm, maximum
    Starting voltage: Initial value ±10 volts
    Plateau length: 150 volts, minimum
    Relative plateau slope: 0.25 percent/Vdc, maximum
    End-of-plateau voltage: 760 volts, minimum
    Response (count rate) (1): Initial value ±10 percent
    Pulse amplitude (1) and (2): Initial limits.

11. The following tests shall be performed at the beginning and at the conclusion of the holding period: Photosensitivity, background and contamination, starting voltage, response (count rate) (1), and pulse amplitude (1) and (2), except that starting voltage shall not vary more than 10 volts.

12. The tube shall be mounted with its longitudinal axis horizontal and perpendicular to the direction of motion by means of a rigid clamp at the center of the cathode.

13. Method B (test method 6216); the scaler of note 5 may be utilized for this test.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Navy -
Air Force - 11
DSA - ES

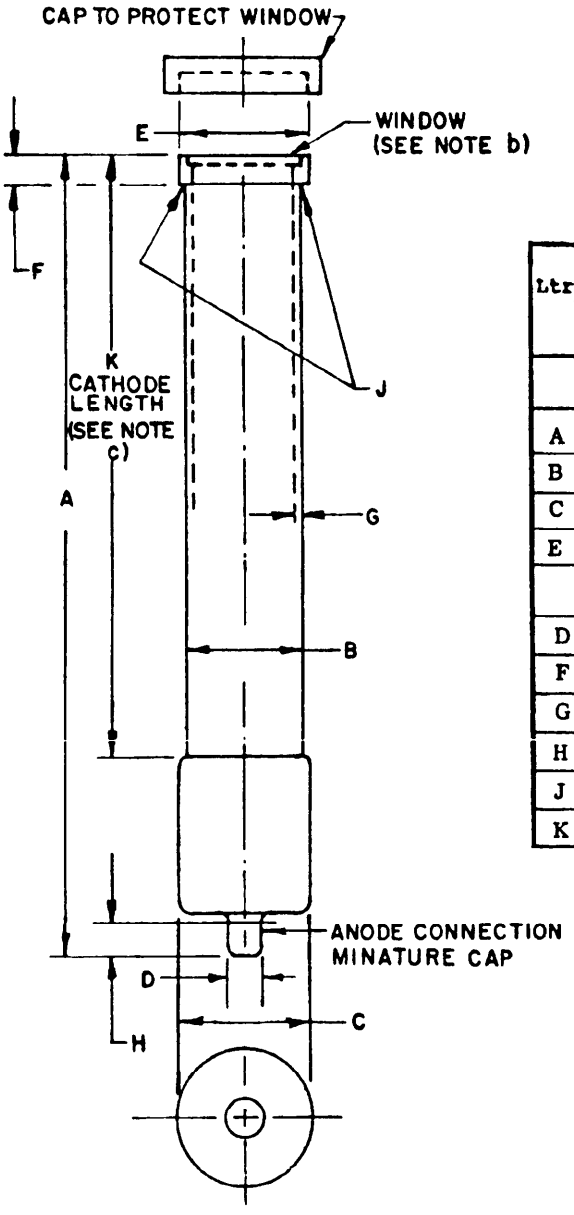
User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force -

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2456)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.750(146.05)	6.000(152.40)
B	.860(21.84)	.890(22.61)
C	---	1.000(25.40)
E	.985(25.02)	1.015(25.78)
Quality conformance inspection, part 3 (periodic-check)		
D	.245(6.22)	.255(6.48)
F	.150(3.81)	.225(5.72)
G	.047(1.19)	.078(1.98)
H	.250(6.35)	---
J	89°	91°
K	4.250(107.95)	4.500(114.30)

NOTES:
a. The envelope will not have an exposed seal-off tip.
b. Mica window 3.0 to 4.0 milligrams per cm2.
c. Cathode: Allegheny Ludlum Sealmet 1 Steel for glass to metal seals, or equal.

FIGURE 1. Outline drawing of electron tube type 5979.

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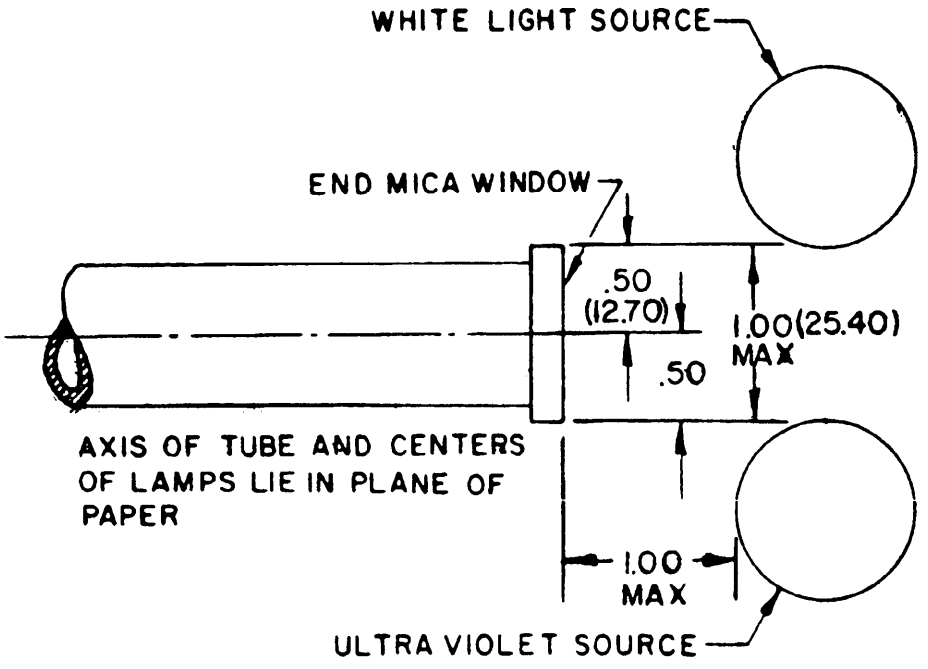


FIGURE 2. Position of tube for photosensitivity test.

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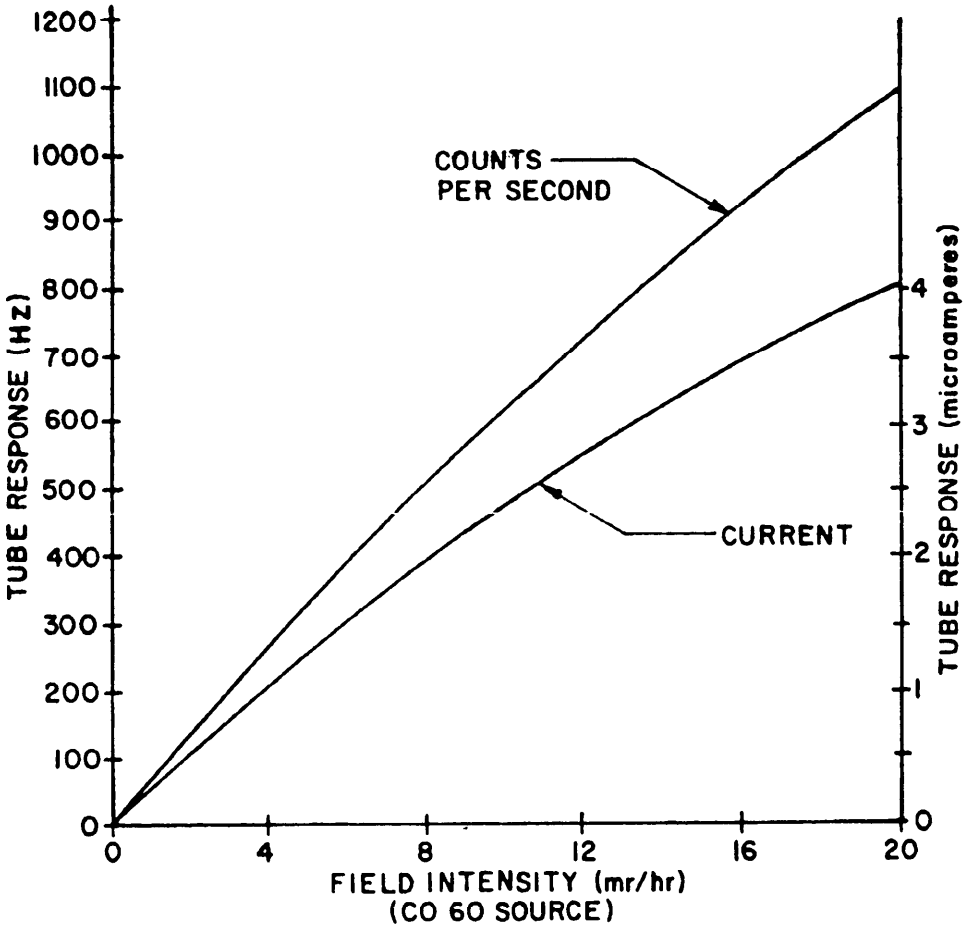


FIGURE 3. Response characteristics of typical type 5979 counter tube.

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specified in 4.4.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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electric wire having an insulating covering comprising a cross- features